Daily Epsilon

My solutions to the daily maths problems from your Daily Epsilon of Math calender.

Day 7

$$-\frac{4}{3}(x-25) = \frac{3}{4}(x+25)$$

Solution

Multiply both sides by 12;

$$\cancel{12} \cdot -\frac{4}{\cancel{3}}(x-25) = \cancel{12} \cdot \frac{3}{\cancel{4}}(x+25)$$

$$4 \cdot -4(x-25) = 3 \cdot 3(x+25)$$

$$-16(x-25) = 9(x+25)$$

$$-16x + 400 = 9x + 225$$

$$-16x - 9x = 225 - 400$$

$$-25x = -175$$

$$x = 7$$

Day 9

Find the number of digits in;

$$14^{22} \cdot 7^{-}20$$

Solution

Figure 1: A screenshot of a toot of mine that i made on mastodon.

Of course, stupid me didn't read the instructions. I just saw the the figures and i dived strsight into solving. i got to a point and i wondered to myself that this doesn't make any sense at all. I then checked and saw that i was supposed to find the number of digits in the *equation.*
A little simplification before jumping into finding the number of digits;

$$(7 \cdot 2)^{22} \cdot 7^{-}20$$

$$(7^{22} \cdot 2^{22}) \cdot 7^{-}20$$

$$\text{recall that : } \; a^b + a^c = a^{b+c}$$

$$\therefore (7^{20} \cdot 7^2 \cdot 2^{22}) \cdot 7^{-}20$$

$$\text{recall that : } \; \left(a^{-}1 = \frac{1}{a}\right)$$

$$\therefore (7^{20} \cdot 7^2 \cdot 2^{22}) \cdot \left(\frac{1}{7^{20}}\right)$$

$$(\cancel{7^{20}} \cdot 7^2 \cdot 2^{22}) \cdot \left(\frac{1}{\cancel{7^{20}}}\right)$$

our simplification finally brings us here;

$$= 7^2 \cdot 2^{22}$$

Now, to find the length of the result, the only way i know how to do this is by writing some code that computes the powers of these integers and then prints the length of the result. My language of choice here is java. Yes, java. python, you say? nah.

Here goes the java code;

```java
public class Epsilon {
    public static void main(String[] args) {
        int a = (int) Math.pow(7, 2);
        int b = (int) Math.pow(2, 22);

        int c = a * b;

        System.out.println(String.valueOf(c).length());
    }
}
```

I could inline the above code but it might make it difficult to look at than it already is if you are not familiar with the java programming language or programming in general.[1] When you run this in an Integrated Development Environment(IDE) that supports java, the result you get is 9.

$$\therefore \text{ the length of the result of } \ 7^2 \cdot 2^{22} \ \text{ is } 9.$$

Day 14

$$\frac{95+87}{8+5}$$

Solution

i saw this problem and thought to myself, there had to be some catch that i am not aware of but there isn't actually. That's the question. Simplify, basically. So this is pretty straightforward, you have to sum the numbers in the numerator as well as the denominator.

$$\frac{95+87}{8+5} = \frac{182}{13} = 14$$

Day 16

find the number of integers that satisfy;

$$\frac{5}{y} \leq -0.3$$

Solution

$$\frac{y}{5} \leq -\frac{1}{0.3}$$

$$y \leq -\frac{5}{0.3}$$

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[1] inlining is basically cleaning your code where instead of having multiple lines, you can just have a single line like this;

```java
public class Epsilon {
    public static void main(String[] args) {
        System.out.println(String.valueOf(Math.pow(7, 2) * Math.pow(2, 22)).length());
    }
}
```

$$y \leq -16.67$$

What is an integer? An integer is a whole number, meaning that it isn't a fraction. 0, −1, 2 are integers. 3.14, 2.713 aren't. From this we can already tell that −16.67 isn't the answer we are looking for but rather, the number of whole numbers that satisfy the equation.
Take ***−16***: put it in the original equation and you have;

$$-0.31 \leq -0.3$$

which is essentially equal to the each other. Ergo, −16 satisfies the equation.
Take ***−15***: repeat the same process from above and you have;

$$-0.33 \leq -0.3$$

which is true so −15 satisfies the equation.
Continuing to do this until you get to −1.
Take ***−1***:

$$-5 \leq -0.3$$

which is also true. Take ***0***:

$$0 \leq -0.3$$

This is false From −16 to −1, that is 16 integers so there are just 16 integers that satisfy the original equation.

Day 17

$$x > 0$$

$$\sqrt{21-x} + \sqrt{x+32} = 9$$

Solution

The first instinct here is to square both sides of course.

$$\left(\sqrt{21-x} + \sqrt{x+32}\right)^2 = 9^2$$

Expanding the *R.H.S* of the above equation yields;[2]

$$(21-x) + 2\sqrt{(21-x)(x+32)} + (x+32) = 81$$

$$21 - \cancel{x} + \cancel{x} + 32 + 2\sqrt{(21-x)(x+32)} = 81$$

$$53 + 2\sqrt{(21-x)(x+32)} = 81$$

$$\cancel{2}\sqrt{(21-x)(x+32)} = \cancel{28}$$

[2]I made a big mistake here when i attempted this problem on paper. I forgot that there were two duplicate terms that should be together i.e.

$$\sqrt{(21-x)(x+32)}$$

. It has been so long i did any serious math that i just made such serious error.

$$\sqrt{(21-x)(x+32)} = 14$$

squaring both sides;

$$(21-x)(x+32) = 196$$

Expanding the brackets;

$$672 - 11x - x^2 = 196$$

$$x^2 + 11x - 476 = 0$$

Using the Wolfram calculator;[3]

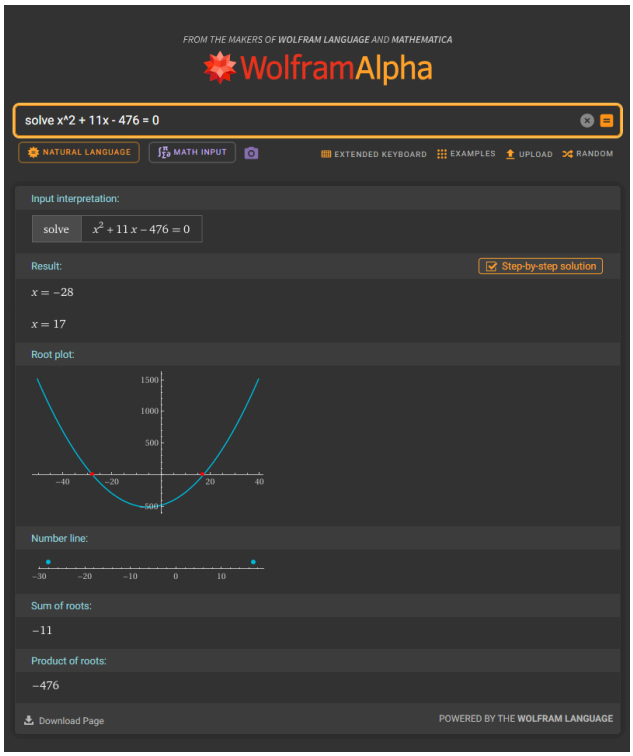

Figure 2: A screenshot of the results using Wolfram

Looking at Figure 2, we can see the results are $x = -28, 17$

since $x > 0$ therefore $x = 17$.

Day 20

The product of two rolls of a six-sided die is greater than 8 with a probability of $\frac{x}{36}$

Solution

Well, this is pretty straightforward, I believe. You have a single six-sided die and you roll it twice. With each roll of the dice, there are six possible outcomes so to find the possible products of the outcomes of the two rolls, we can make use of a table(6x6).

[3] At this point, I decided not to *waste* my time by doing the factorization looking for factors of 476 whose sums/differences resulted in −11 so i pulled up Wolfram's calculator

	1	2	3	4	5	6
1	1	2	3	4	5	6
2	2	4	6	8	**10**	**12**
3	3	6	**9**	**12**	**15**	**18**
4	4	8	**12**	**16**	**20**	**24**
5	5	**10**	**15**	**20**	**25**	**30**
6	6	**12**	**18**	**24**	**30**	**36**

Table 1: Table showing the products of the results of two rolls of a six-sided die.

Counting those products that are greater than 8. There are 20 of them. So $x = 20$

$\therefore$ the probability of the product of two rolls of a six-sided die being greater than 8 is $\frac{20}{36}$.

Day 22

$$+7 \cdot 10$$
$$-4 \cdot 8$$
$$+10 \cdot 4$$
$$-8 \cdot 7$$

Solution

Yup, you read that problem correctly. This was exactly the problem for the 22nd of March. This is pretty straightforward. So what we should do is bring out BODMAS[4] and according to this rule, we should be handling all the multiplications before adding or subtracting. So doing this results in

$$\begin{aligned} &= +70 - 32 + 40 - 56 \\ &= 110 - 88 \\ &= 22 \end{aligned}$$

[4]If this is your first time seeing this, allow me to explain the usefulness of Brackets, Of, Division, Multiplication, Addition, Subtraction. So

$$a + b \cdot c + d$$

gives you a different result if you add the terms together before multiplying them than if you multiplied terms together before then adding up. So to prevent from this variation in result, we have this in place. So you have to take the operations in this order. Why? I still do not know. I remember asking myself this question when i was teaching kids math during my service year. I should fix up on that.

Day 23

$23^{29} \bmod 29$

Solution

This is quite the problem. lol.[5] I am currently struggling to solve this problem despite the approaches that i have taken.[6]
The first thought i had was writing a simple java program that found the value of 23 to the power of 29 using the function pow( ) imported from the Math class and then using the modulus operator to find the final answer but that didn't work because the data type i used for storing these values was 'int' and there is a maximum value that can be stored in int which is

$$2^{31}$$

and this is clearly smaller than 23 to the 29th power. I thought of using BigInteger but there is a problem in trying to use the modulus operator with the BigInteger variable.[7]
I tried using the modulus function for BigInteger but yet i still am not getting the answer which leaves me with the only other option of trying to breakdown the number into smaller bits that i can then perform computation on but 2 is clearly a prime which sucks. I might also try using the commutative property of modular aithmetic to break it down. I don't know. Lemme keep on trying.
Several minutes later, and i have been rescued by the commutative power of modular arithmetic as well as a small java program that i wrote.

> An important property of the remainder is that in addition, subtraction and multiplication, the remainder can be taken before the operation:
>
> $$\begin{aligned}(a+b) \bmod m &= (a \bmod m + b \bmod m) \bmod m \\ (a-b) \bmod m &= (a \bmod m - b \bmod m) \bmod m \\ (a \cdot b) \bmod m &= (a \bmod m \cdot b \bmod m) \bmod m\end{aligned}$$
>
> Thus, we can take the remainder after every operation and the numbers will never become too large.

Figure 3: A screenshot of an excerpt from the Competitive Programmer's Handbook written by Antti Laaksonen. (*pp 6-7*)

So just to refresh my knowledge of modular arithmetic, i consulted the Competitive Programmer's Handbook written by Antti Laaksonen and with that i just knew what to do. First, i had to break them down(ignoring the modulus for now);

$$23^{29}$$

$$\begin{aligned}&= 23^{19} + 10 \\ &= 23^{19} \cdot 23^{10} \\ &= 23^{9} \cdot 23^{10} \cdot 23^{5} \cdot 23^{5} \\ &= 23^{5} \cdot 23^{4} \cdot 23^{5} \cdot 23^{5} \cdot 23^{5} \cdot 23^{5}\end{aligned}$$

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[5] Trying to also document my struggles with problems and the approaches that clearly didn't work.

[6] As i was writing this sentence, i suddenly had an a possible insight for an approach on how to go about solving the problem.

[7] If i haven't mentioned this or you are unfamiliar, there is an obvious clue to help you with solving the daily problems and that is the fact that the answer to the problem is that day's date. So going by this, The answer for today should be 23 and nothing else.

Now, bringing in the modulus part of the equation and using our updated knowledge on modular arithmetic from Figure 3.

$$= (23^5 \cdot 23^4 \cdot 23^5 \cdot 23^5 \cdot 23^5 \cdot 23^5) \bmod 29$$

$$= (23^5 \bmod 29 \cdot 23^4 \bmod 29 \cdot 23^5 \bmod 29 \cdot 23^5 \bmod 29 \cdot 23^5 \bmod 29 \cdot 23^5 \bmod 29) \bmod 29$$

We can now write a program that looks for the fifth and fourth power of 23 which would then find the modulus;

```
public class Day23 {

    public static void main(String[] args) {

        //finding the fourth and fifth power of 23 which then casts our answer to
type int because the type of the result of the pow() method is double.
        int h = (int) Math.pow(23, 5);
        int h1 = (int) Math.pow(23, 4);

        //finds the modulus of the results stored in the above variable.
        int i = h % 29;
        int i1 = h1 % 29;

        //multiplies the results.
        int c2 = (int)Math.pow(i, 5) * i1;

        //prints the result of c2 mod 29.
        System.out.println("the result of 23^29 mod 29 is " + c2%29);

    }
}
```

When you run this program, the printed statement on the terminal/console is "the result of 23^29 mod 29 is 23". And there it is–our answer–23.

Day 24

A basketball team made 25% of its 3-point shots and 44% of its 2-point shots to score 40 points from 49 shots. How many 3-point shots were attempted?

Solution

After reading this question hundreds and a page filled with terrible attempts at a solution, i can finally say i have solved it and all i needed was a piss break. I took a piss break, came back and started afresh. Now to the solution.
49 shots were made consisting of 3 points shot and 2 point shots. Let

$$x = \text{amount of 3pts shot attempted.}$$
$$y = \text{amount of 2pts shot attempted.}$$

Given this, we can develop the following equation knowing the total shots made which is 49;

$$x + y = 49$$

Of the total amount of 3pts shots attempted, only 25% went in and of the total amount of 2pts shots attempted, only 44% went in. The total points scored from the 49 shots is 40 points which would mean that just 25% of the 3 points shots attempted summed together with 44% of the 2 points shots attempted result in a total score of 40. Given this, we can develop the following equation;

$$25\%(3x) + 44\%(2y) = 40$$

$$0.25(3x) + 0.44(2y) = 40$$

$$0.75x + 0.44y = 40$$

We now have a set of simultaneous equations.

$$x + y = 49 \rightarrow \text{eqn 1}$$

$$0.75x + 0.44y = 40 \rightarrow \text{eqn 2}$$

We will solve for just x since that is what we are concerned with. So making y the subject of formula in eqn 1 and substituting for y in eqn 2.

From eqn 1, we have $y = 49 - x$ and then substituting for y in eqn 2

$$0.75x + 0.88(49 - x) = 40$$

$$0.75x + 43.12 - 0.88x = 40$$

$$0.75x - 0.88x = 40 - 43.12$$

$$\cancel{-}0.13x = \cancel{-}3.12$$

$$x = \frac{3.12}{0.13}$$

$$x = 24$$

$\therefore$ The amount of 3-points shots attempted was 24.

Day 26

$$\sum_{n=1}^{6} \tan^2\left(\frac{n\pi}{14}\right)$$

Solution

$$\begin{aligned}\sum_{n=1}^{6} \tan^2\left(\frac{n\pi}{14}\right) &= \tan^2\left(\frac{\pi}{14}\right) + \tan^2\left(\frac{\cancel{2}\pi}{\cancel{14}}\right) + \tan^2\left(\frac{3\pi}{14}\right) + \tan^2\left(\frac{\cancel{4}\pi}{\cancel{14}}\right) + \tan^2\left(\frac{5\pi}{14}\right) + \tan^2\left(\frac{\cancel{6}\pi}{\cancel{14}}\right) \\ &= \tan^2\left(\frac{\pi}{14}\right) + \tan^2\left(\frac{\pi}{7}\right) + \tan^2\left(\frac{3\pi}{14}\right) + \tan^2\left(\frac{2\pi}{7}\right) + \tan^2\left(\frac{5\pi}{14}\right) + \tan^2\left(\frac{3\pi}{7}\right)\end{aligned}$$

At this point, i used the calculator of my phone to compute the values and for the first three terms of the equation, i noticed that the results were too low that if i were to sum them all up, i wouldn't get

the result i wanted as the clue suggests.[8] So off to Wolfram, I went.

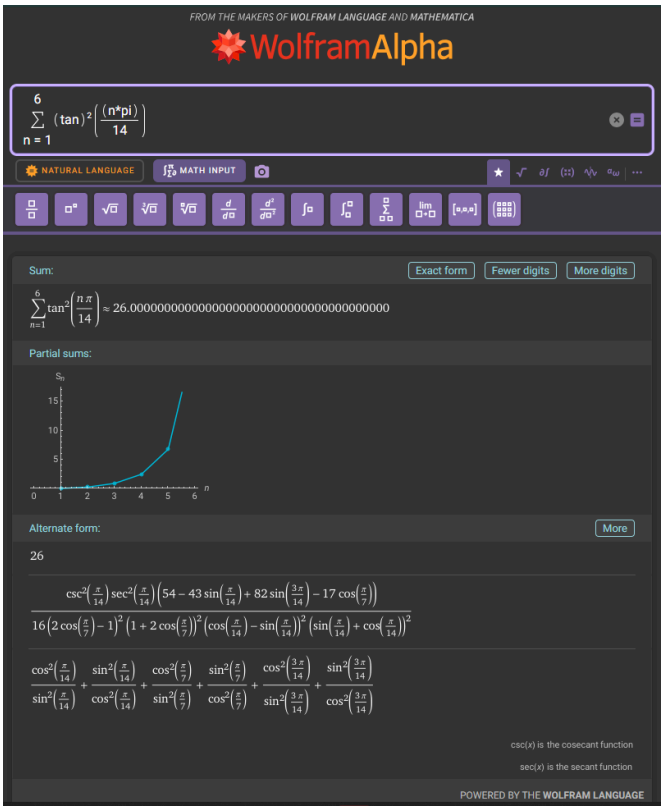

Figure 4: A screenshot of the results using Wolfram.

From the diagram, we can see that the result is 26.

$$\therefore \sum_{n=1}^{6} \tan^2\left(\frac{n\pi}{14}\right) = 26$$

[8] or maybe i didn't write the expression well in the calculator.

The order in which these problems were solved.

- Day 14.
- Day 9.
- Day 17.
- Day 16.
- Day 7.
- Day 20.
- Day 22.
- Day 23.
- Day 24.
- Day 26.